

Langmuir probes in a wide range of plasma density: Emissive and non-emissive probes

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ABSTRACT

Ion saturation current and floating potential were calculated for non-emissive and emissive probes using a kinetic approach. A wide range of plasma pressures was considered: from almost collisionless $R/l \ll 1$ up to almost continuous ($R/l \gg 1$) cases, where R is the probe radius and l is the ion free path. Calculations were performed for cold ions (ion temperature $T_i = 0$) and for warm ions (the ratio of the ion temperature to the electron temperature T_i/T_e not exceeding approximately 0.3). The Debye radius was assumed to be much narrower than all of the characteristic lengths. Simple approximation formulas are suggested for the ion current and the floating potential for both (non-emissive and emissive) types of probes. Calculations showed that the ion velocity at the sheath-presheath boundary corresponds to the Bohm criterion for all the considered cases.

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I. INTRODUCTION

Introduced by Mott-Smith and Langmuir,¹ electric probes remain a powerful tool in plasma diagnostics. Depending on the probe bias, they provide information on either plasma electrons at the electron part of the current–voltage (I – V) characteristic or ions at the ion part of the I – V characteristic. The electron part is used to measure electron temperature and the electron energy distribution function. The ion branch of the current–voltage characteristic is used to measure the density of charged particles. In this paper, we are focused on the ion branch of the I – V characteristic when the probe is negatively biased. In this mode, electrons are trapped in the probe's proximity. Their current to the probe is limited by the retarding potential. The flow of ions, by contrast, is accelerated toward the probe. For stable probe operation, at the sheath-neutral plasma boundary, ions should have an average velocity equal to or, in excess of, the local speed of sound (the Bohm criterion). How the ions gain this velocity depends on the relation between the ion free path l and the probe radius R . The extremes are: the collisionless regime $R \ll l$ and the collision dominated (diffusive) mode $R \gg l$. Theories of both regimes are relatively well developed: collisionless mode—by Bernstein and Rabinowitz,² Laframbois,³ and Allen *et al.*;⁴ collision dominated—the works by Su and Lam,⁵ Su and Kiel,⁶ and Benilov and Rogov.⁷ There are a number of books and journal reviews that describe different probe types and methods of interpretation of their signals. To name just a few, these classic

reviews^{8–12} and more recent comprehensive ones^{13,14} should be mentioned.

Between collisionless and collision dominated extremes, there is rather a wide intermediate region. To see it, let us estimate the ion free path for argon plasma. Consider a probe of $R = 100 \mu\text{m}$ radius. An argon ion cross section of charge exchange collisions is $S \sim 10^{-28} \text{ m}^2$.¹⁵ At the plasma density of $N \sim (\text{SR})^{-1} \sim 10^{22} \text{ m}^{-3}$, the ion free path is equal to the probe radius. With argon gas at room temperature, it corresponds to a pressure of 40 Pa. At substantially lower pressure, the probe works in the collisionless mode and at higher pressure, in the collision dominated mode. Theory to calculate the ion current to the probe in this pressure range practically does not exist. One goal of this paper is to fill this niche.

Another goal of this work deals with ion current to emissive (electron emitting) probes. Electron emission by this type of probe could be thermionic emission or secondary electron emission under ion impact or photo-effect produced by a laser. Emitted electrons add negative electric charge to the sheath, thus decreasing the potential drop inside it. Although emissive probes were proposed at almost the same time as the non-emissive probes,¹⁶ for a long time they did not attract as much attention.

The first experiment with the emissive probes was performed by Kemp and Sellen.¹⁷ They noticed that an increase in the probe emission first gave rise to the float potential (it becomes less negative), but

after some critical value of the emission, the float potential stopped growing and saturated. It was concluded that this event corresponds to equality of the plasma and the probe potentials. Solution of the Poisson equation inside the sheath confirmed their conclusion. Almost simultaneously, Hobbs and Wesson published a paper where they considered plasma parameters close to an emitting wall (plane probe).¹⁸ They showed that the floating potential shrinks as the electron emission intensifies. The critical change in the probe operation takes place as the emission current approaches the electron current provided by the plasma. At that moment, the probe potential becomes lower than the potential of the undisturbed plasma by approximately T/e . The low value of this voltage drop allows one to measure the plasma potential by measuring the floating potential. At higher emission currents, the field changes the sign, and a potential well is created. Further emission could lead to inverting potential in the whole sheath.¹⁹

Different techniques to measure plasma potential were suggested: floating potential at high emission, a separation technique, and an inflection point at zero emission.²⁰ Although according to Ref. 21, the inflection point is the most accurate, and the floating potential still remains one of the most popular in view of its relative easiness and reasonable accuracy.^{22–24} In this paper, we focus mainly on this technique.

In the vast majority of the papers on emission probes, the collisionless regime has been suggested. We are aware of only work²⁵ where ion collisions inside the sheath and presheath were taken into account. As in the case of non-emissive probes, the intermediate situation $R \sim l$ is not covered. An attempt to fill this gap for emissive probes is another purpose of this work.

There is another niche in emissive probe simulation. Almost all the models consider the 1D (plane). We are aware of only two papers that deal with cylindrical probes. One of them is already mentioned,²⁵ another²⁶ treats collisionless probes. Other than these two papers, we are not aware of neither spherical nor of cylindrical emissive probe analysis.

This paper is organized as follows: In Sec. II, non-emissive spherical and cylindrical probes are considered assuming the cold ion approximation. In Sec. III, the same is done for emissive probes. In Appendix A, details of the numerical procedure are explained. In Appendix B, we dropped the cold ion approximation. Correction for spherical probes with “warm” ions (non-zero ion temperature $T_i/T_e < \sim 0.3$) is presented.

II. NON-EMISSIVE PROBES. INTRODUCTION

There are two applications of electric probes we bear in mind. One is plasma density measurement using ion saturation current. When the probe is negatively biased, the ion current to the probe (let us say, spherical one) is

$$I_i = 4\pi e N_0 R^2 \sqrt{\frac{2T_e}{M}} \times A(R/l), \quad (1)$$

where $A(R/l)$ is a function depending on the R/l ratio, the function we are interested in. Here, N_0 (m^{-3}) is the undisturbed plasma density; T_e (eV) is the electron temperature. M is the ion mass, and e is the elementary charge. Since the electron current is suppressed at a negative enough potential, the ion saturation current is practically the only current to the probe. According to (1), it allows one to measure the plasma density.

The second application is related to measuring plasma potential by means of the floating probe. At some probe potential (float potential), there is the condition of no current to the probe. The electron and the ion currents balance each other. Assuming that almost all the potential difference between the undisturbed plasma and the probe takes place inside the sheath and knowing its value²⁷

$$V_{\text{sheath}} = -\frac{T_e}{2e} \ln \left[\frac{2\pi m}{M} \right], \quad (2)$$

one may judge the local plasma potential. Here, V is the potential measured in Volts, and m is the electron mass. Note, however, that in addition to the potential drop inside the sheath, there is a potential drop in the quasi-neutral plasma V_{plasma} (presheath region), see Fig. 1.

In the case of a collisionless probe, the value of the V_{plasma} is known and can be easily taken into account. However, its value increases with R/l increase. Hence, it should be included in consideration when interpreting the experimental data. Below, we present the results of our calculation of V_{plasma} and of ion saturation current in a wide range of R/l ratios. Partially, it is based on the author's old publications.^{28,29}

A. Non-emissive probes: Assumptions

We consider plasma with the cold ions so that the motion of the particles is strictly radial. The concept of Orbital motion³⁰ is, therefore, not relevant. The plasma is assumed to be dense enough so that the Debye length is short compared to R and l . For example, for the electron density $3 \times 10^{17} \text{ m}^{-3}$ and $T_e \sim 1 \text{ eV}$, r_D becomes equal to the probe radius of $100 \mu\text{m}$. At higher electron density, the sheath becomes narrower than the probe radius. In this paper, we consider only the quasi-neutral plasma, assuming that $r_D \ll R$ and $r_D \ll l$. At the limit of $r_D/l \rightarrow 0$, the quasi-neutral plasma is strictly separated from the sheath with infinite electric field at the boundary. At this limit, the V_{sheath} can be calculated separately. How this problem is tackled numerically is explained in detail in the Sec. IIB 1 and Fig. 2. The problem of plasma-sheath boundary at non-zero r_D/l has been considered in a series of works by Godyak and Sternberg.^{31–33}

B. Spherical probe: Equations

First, we present formulas for a spherical probe. The necessary corrections for a cylindrical probe will be given later.

The ruling equation is equality of ion and electron densities in quasi-neutral plasma. Let us start with the formula for the ion density. The main mechanism of ion-atom collisions is the charge exchange in view of its high cross section. In the cold ion limit, ions lose all their energy accumulated in the electric field and after collision they start their next free path with zero velocity. Ion density at some radius r is the sum of the participation of all the ions that experienced their last collisions at different radii $r_1 > r$.

So, if the previous collision took place at r_1 , at location r the ion has the velocity (m/s)

$$v(r, r_1) = \sqrt{\frac{2e}{M} [V(r_1) - V(r)]}. \quad (3)$$

The probability for the ion to travel from r_1 to r without collisions is $\exp(r/l - r_1/l)$. By integrating over all the possible locations of the previous collisions, one has for the ion density at radius r (spherical case)

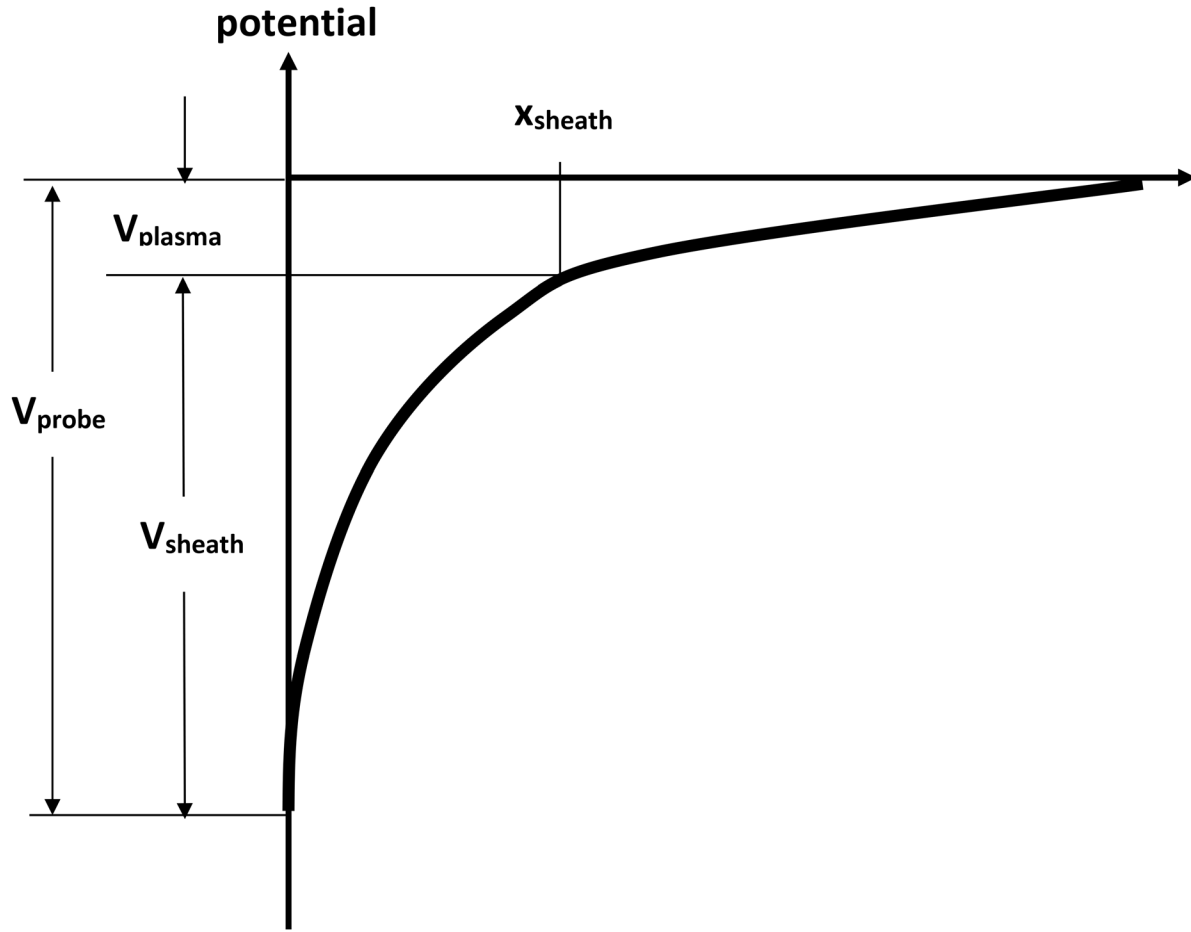


FIG. 1. Schematic of the potential distribution around the probe. Potential of the undisturbed plasma is assumed to be zero.

$$n_i = \frac{I_i}{4\pi r^2} \int_r^\infty \frac{\exp\left(\frac{r}{l} - \frac{r_1}{l}\right) dr}{v(r, r_1) \frac{dr}{l}}, \quad (4)$$

where I_i is the ion current. As for electrons, they are trapped by the electric field. Dependence of their density on potential is described by the Boltzmann factor, i.e., $n_e = N_0 \exp(eV/T)$, where N_0 is the plasma density far from the probe (Hereafter except [Appendix B](#), by T we mean the electron temperature.). The equation of electrical neutrality, for a spherical probe, is therefore

$$\exp\left[\frac{eV(r)}{T}\right] = \frac{I_i}{4\pi e N_0 r^2} \times \int_r^\infty \frac{\exp[(r - r_1)/l] dr_1}{\sqrt{\frac{2e}{M} [V(r_1) - V(r)]}}. \quad (5)$$

For a cylindrical probe, this formula takes the form

$$\exp\left[\frac{eV(r)}{T}\right] = \frac{I_i}{2\pi e N_0 L r} \times \int_r^\infty \frac{\exp[(r - r_1)/l] dr_1}{\sqrt{\frac{2e}{M} [V(r_1) - V(r)]}}, \quad (6)$$

where L is the probe length.

1. Collisionless mode

In this subsection, we briefly remind the reader of the well-known results for collisionless regimes of cylindrical and spherical probes ($R \ll l$). This case has been considered in detail by Allen *et al.* in 1957.⁴

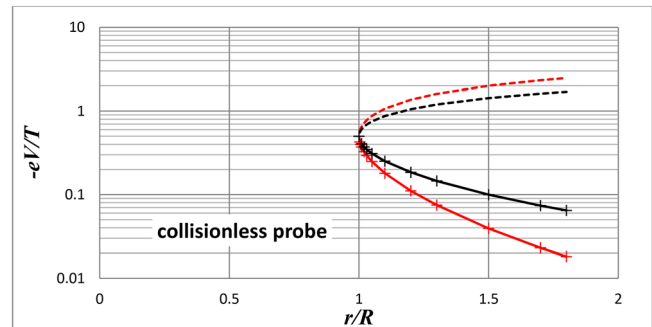


FIG. 2. Calculated radial distribution of potential in quasi-neutral plasma for the collisionless spherical (red) and cylindrical (black) cases. Note that condition $dV/dr \rightarrow \infty$ takes place at $eV/T = -1/2$. Upper branches solutions of Eq. (8) are of no physical content and given for symmetry.

When treating the collisionless limit, it is convenient to introduce the dimensionless parameters (spherical probe)

$$\rho = \frac{r}{R} \quad \phi = \frac{eV}{T} \quad A \equiv \frac{I_i}{4\pi n_0 R^2} \sqrt{\frac{M}{2T}}. \quad (7)$$

In the collisionless limit, $\phi(r_1) \ll \phi(r)$ and integral in (5) can be easily calculated. We have

$$e^\phi = \frac{A}{\rho^2 \sqrt{-\phi}}. \quad (8)$$

Parameter A should be chosen in such a way that the plasma-sheath boundary condition the place at $\rho = 1$. For the given dimensionless radius and parameter A value, (8) is the equation for $\phi(\rho)$.

The results of the numerical solution are shown in Fig. 2, where the spatial distribution of the plasma potential is presented.

Figure 2 clearly shows what takes place at the boundary. The closer the calculations come to the boundary, the electric field (dV/dr) rises more and more sharply. Finally, at some radius, there is no solution for a smaller radius. Of course, numerically, there is some uncertainty in the value of the field as it approaches the boundary. However, there is absolutely negligible error in the values of the potential and the position of the boundary.

As expected, the plasma potential is equal to $-T/2e$. The parameter A is equal to $\frac{\exp(-\frac{1}{2})}{\sqrt{2}} = 0.429$ at the neutral plasma-sheath boundary for both the (cylindrical and spherical) probes. This result can be obtained analytically.⁴

2. Collisional mode (spherical probe)

Now, let us consider the case of an arbitrary R/l ratio. Different non-dimensionality is used in this section. Now, radius is measured in the ion free path: $\rho = r/l$, and for dimensionless ion current, we introduce the parameter

$$B = A \left(\frac{R}{l} \right)^2 = \frac{I_i}{4\pi e n_0 l^2} \sqrt{\frac{M}{2T}}. \quad (9)$$

The corresponding equation is

$$\exp(\phi) = \frac{B}{\rho^2} \int_{\rho}^{\infty} \frac{\exp(\rho - \rho_1)}{\sqrt{\phi_1 - \phi}} d\rho_1. \quad (10)$$

The equation is solved starting from a very large dimensionless radius, where the approximation of low electric field is valid, down to some point where $d\phi/d\rho$ turns to infinity (actually, becomes very large). This value of ρ is considered equal to the probe radius. For each value of the parameter B, there is a corresponding probe radius. Thus, changing B over a wide range, we obtained the probe characteristics for any desired R/l value. The numerical procedure to solve this equation is explained in Appendix A.

As the collisionless limit was considered above, now let us proceed to the opposite limit. For the continuum case, $R/l \rightarrow \infty$, the solution can be obtained analytically. In this extreme, potential doesn't change much at a distance of an ion free path.

Hence, one can put $\phi_1 - \phi \approx \frac{d\phi}{d\rho}(\rho_1 - \rho)$. Now, the condition of neutrality yields

$$\exp(\phi) = \frac{B}{\rho^2} \left(\frac{d\phi}{d\rho} \right)^{-\frac{1}{2}} \int_0^{\infty} \frac{\exp(-x)}{\sqrt{x}} dx = \frac{B\pi}{2\rho^2} \left(\frac{d\phi}{d\rho} \right)^{-\frac{1}{2}}. \quad (11)$$

This leads to the differential equation

$$e^{2\phi} d\phi = (\pi B)^2 \frac{d\rho}{4\rho^4}. \quad (12)$$

Keeping in mind that $\phi(\infty) = 0$, integration from given radius to infinity yields

$$\phi = \frac{1}{2} \ln \left[1 - \frac{1}{6} (B\pi)^2 \frac{1}{\rho^3} \right]. \quad (13)$$

The condition that points to the plasma-sheath boundary, $\frac{d\phi}{d\rho} \rightarrow \infty$, takes place when B is equal to $\frac{1}{\pi} \left[6 \left(\frac{R}{l} \right)^3 \right]^{1/2}$.

We conclude that $A = B(R/l)^{-2}$ decreases at large probes as

$$A \sim \frac{\sqrt{6}}{\pi} \left(\frac{R}{l} \right)^{-1/2}. \quad (14)$$

3. Results

Figures 3 and 4 show the dimensionless ion current A and the potential drop inside the neutral plasma V_{plasma} for spherical probe.

In the range from $R/l = 0.01$ to $R/l = 100$, the empirical formula

$$A = \frac{0.43}{1 + 0.18(R/l)^{0.85}} \quad (15)$$

(red line in Fig. 3) describes our calculations with accuracy better than 96%.

Another important parameter is the floating potential (potential at no current to the probe)

$$I_i = A(R/l) \times 4\pi e n_0 R^2 \sqrt{\frac{2T}{M}}, \quad (16)$$

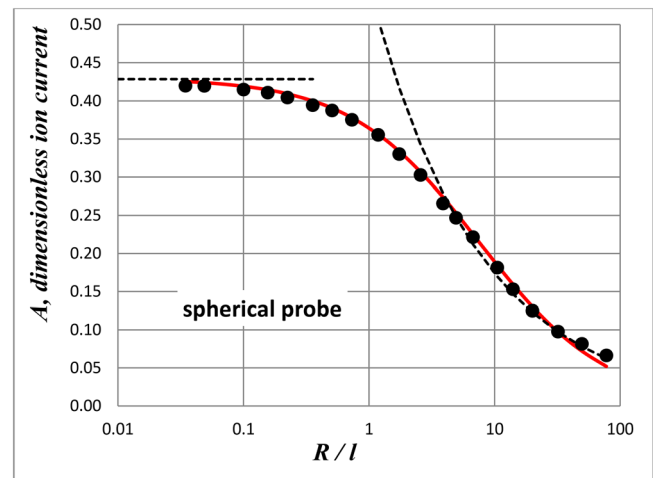


FIG. 3. Dimensionless ion current to the spherical probe. Red line: approximation (15). Dashed lines: solution for extremes (collisionless and continuum cases).

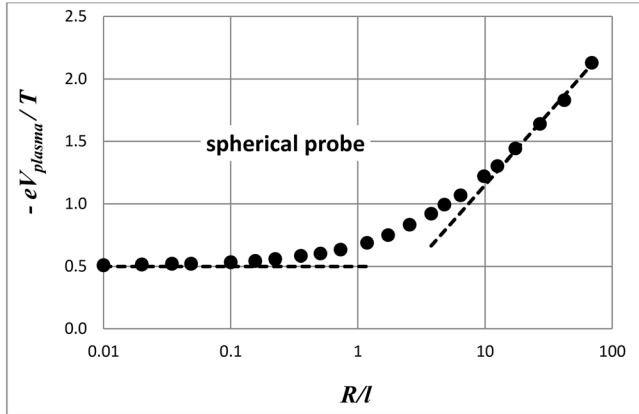


FIG. 4. Voltage drop inside quasi-neutral plasma. Dashed line at high R/l ratio was obtained with formula (19) using asymptotic value of A from (15).

$$I_e = 4\pi e N_0 R^2 \sqrt{\frac{T}{2\pi m}} \exp(\phi_{float}). \quad (17)$$

At the floating condition, these two currents balance each other. For the potential difference ϕ_{float} , one obtains

$$\phi_{float} = \ln A \left(\frac{R}{l} \right) + \frac{1}{2} \ln \frac{2\pi m}{M} + \frac{1}{2} \ln 2. \quad (18)$$

To use the standard formula for the potential drop inside the sheath, i.e., $\frac{1}{2} \ln \frac{2\pi m}{M}$, in (18), we added the term $\ln(\sqrt{2})$ (By redefining formula for parameter A , this additional term can be eliminated.). We see that the potential drop inside the quasi-neutral plasma can be calculated as

$$\phi_{plasma} = \ln \left[\sqrt{2} A \left(\frac{R}{l} \right) \right]. \quad (19)$$

In the collisionless limit, $A = \frac{\exp(-0.5)}{\sqrt{2}}$ and $\phi_{plasma} = -\frac{1}{2}$. The dependence of ϕ_{plasma} on R/l is shown in Fig. 4

C. Collisional probe (cylindrical probe)

For the cylindrical probe, the above formulas take the form

$$B = A \left(\frac{R}{l} \right) = \frac{I_i}{2\pi e n_0 l L} \sqrt{\frac{M}{2T}}. \quad (20)$$

The corresponding equation is

$$\exp(\phi) = \frac{B}{\rho} \int_{\rho}^{\infty} \frac{\exp(\rho - \rho_1)}{\sqrt{\phi_1 - \phi}} d\rho_1. \quad (21)$$

At the collisionless extreme, just as in the spherical case, one has $\phi(\frac{R}{l} \ll 1) = -0.5$ and $A = \frac{\exp(-0.5)}{\sqrt{2}} = 0.43$. At the opposite extreme $R/l \gg 1$, calculations similar to that of (10)–(12) yield $A \sim \frac{\sqrt{2}}{\pi} \left(\frac{R}{l} \right)^{-1/2}$. The results of the numerical solution of the (21) equation are given in Figs. 5 and 6.

For cylindrical probes, we suggest empirical formulas for dimensionless ion current

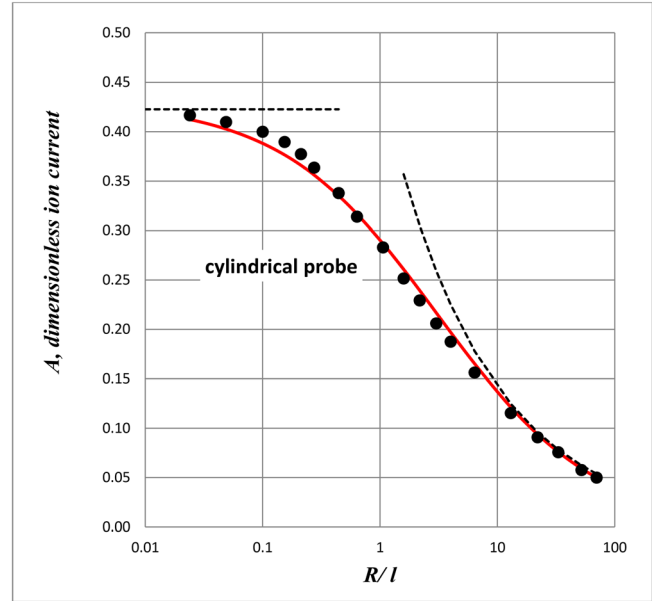


FIG. 5. Dimensionless ion current to the cylindrical probe. Red line: approximation (22). Dashed lines: solution for extremes (collisionless and continuum).

$$A = \frac{0.43}{1 + 0.48 (R/l)^{0.65}} \quad (22)$$

and for potential drop in the case $R/l \gg 1$

$$-e V_{plasma}/T = 0.61 + 0.534 \ln(R/l). \quad (23)$$

D. The ion velocity at the plasma-sheath boundary

In the limit of the collisionless probe, it was known that the ion velocity at the plasma-sheath boundary is equal to $v_{Bohm} = (T/M)^{1/2}$.²⁷ It was not known *a priori* that the same is true for an arbitrary R/l ratio. This condition has been checked. The ion velocity was calculated as a ratio of the ion current density to the plasma density, both at the boundary

$$\frac{v_{boundary}}{v_{Bohm}} = \sqrt{2} A \exp(-\phi_{plasma}). \quad (24)$$

The result of our calculation of the ion velocity is shown in Fig. 7

Substantial scatter of the calculated values of the ion velocity is explained as follows. Close to the plasma-sheath boundary, the potential derivative becomes very large. As a result, the accuracy of the calculations of the potential on a fixed mesh decreases, especially for large values of R/l . Since the ion velocity at the boundary depends on the potential in an exponential way, it leads to some loss in accuracy. Nevertheless, in spite of some scatter, our calculation definitely shows that ion velocity at the plasma-sheath boundary satisfies the Bohm criterion.

III. EMISSIVE PROBE

A. Assumptions

As in the first part, we are interested in the voltage drop inside the quasi-neutral plasma of the emissive probe at the floating mode

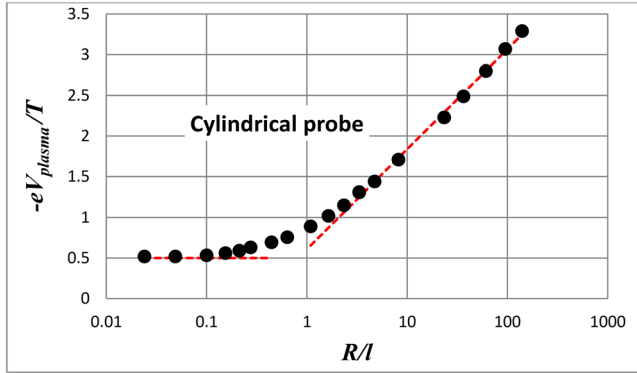


FIG. 6. Potential drop in quasi-neutral plasma adjacent the cylindrical probe. Dashed lines represent the collisionless case and approximation (23).

and in the ion current to the probe. We consider the case of cold ions and cold electrons emitted by the probe so that the motion of both particles is strictly radial. Plasma is assumed to be dense enough so that the Debye radius is short compared to R and l .

The electron-atom collision cross section is much smaller than the ion-atom cross section. This is especially true for plasma of inert gases where there is the Ramsauer minimum for e-a collisions. Thus, for e-Ar collisions, S_{e-Ar} is less than 10^{-20} m^2 (Ref. 34) whereas the cross section for ion charge exchange collisions is on the order of 10^{-18} m^2 .³⁴ In view of this difference, there is a wide range of plasma pressures when electron motion close to the probe is collisionless ($R \ll l_e$) while the ion motion changes from collisionless $R/l \ll 1$ to collision dominated $R/l \gg 1$ (l_e and l are electron and ion free paths, respectively).

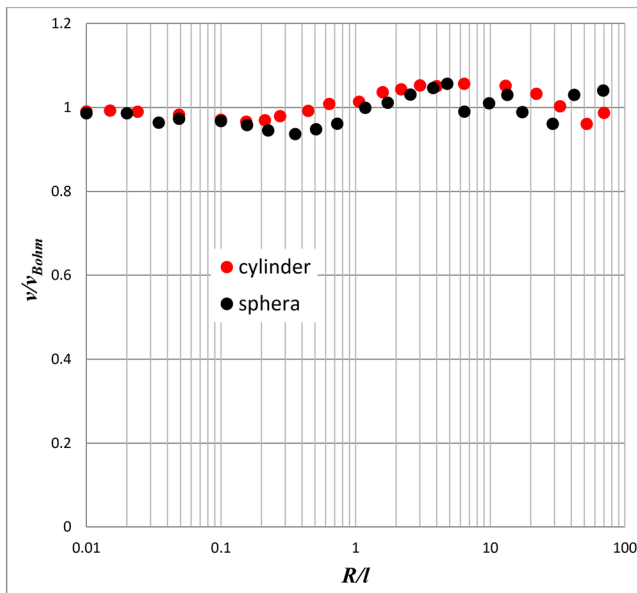


FIG. 7. Ion velocity at the sheath edge to the Bohm velocity ratio.

In this paper, we focus at the presheath. As for the sheath, we assume that the potential distribution inside the sheath is monotonic, thus restricting ourselves to the electron emission current being less than that from the plasma to the probe.¹⁸ The potential drop inside the sheath manifests itself as the initial energy of the emitted electrons at the sheath-plasma boundary and as a part of the total probe-plasma potential difference. The latter is used to find the floating probe potential.

B. Equations

Electron density consists of two parts. One part is that of plasma electrons trapped by the potential barrier. Another is created by the beam of the emitted electrons. The first one is described by the Boltzmann formula. For the density of the emitted electrons N_{emiss} , we have (spherical probe)

$$N_{\text{emiss}} = \frac{I_{\text{emiss}}}{4\pi e r^2} \left[\frac{2e(V - V_{\text{probe}})}{m} \right]^{-1/2}, \quad (25)$$

where I_{emiss} is the electron current emitted by the probe. The initial energy of the emitted electrons, proportional to the probe temperature, is assumed to be zero.

The condition of quasi-neutrality takes the form

$$\exp\left[\frac{eV(r)}{T}\right] + \frac{I_{\text{emiss}}}{4\pi e N_0 r^2} \frac{1}{\sqrt{\frac{2e}{m} [V(r) - V_{\text{probe}}]}} = \frac{I_i}{4\pi e N_0 r^2} \times \int_r^\infty \frac{\exp[(r - r_1)/l]}{\sqrt{\frac{2e}{m} [V(r_1) - V(r)]}} \frac{dr_1}{l}. \quad (26)$$

It is convenient to introduce $g_i = I_{\text{emiss}}/I_i$. For the floating probe, the ratio of the emission current to the current of the plasma electrons flowing to the probe is $g_e = I_{\text{emiss}}/I_{e,\text{plasma}} = g_i/(1 + g_i)$. $I_{e,\text{plasma}}$ is the electron current from the plasma. Parameters g_i and g_e can be used interchangeably. Similar to g_e , g_i is an external independent parameter.

Introducing as before the parameter B [see formula (9)], we obtain the dimensionless equation for potential

$$\exp(\phi) = \frac{B}{\rho^2} \left[\int_\rho^\infty \frac{\exp(\rho - \rho_1)}{\sqrt{\phi_1 - \phi}} d\rho_1 - \frac{g_i \sqrt{m/M}}{\sqrt{\phi - \phi_{\text{probe}}}} \right]. \quad (27)$$

Formally speaking, unlike Eq. (10), Eq. (27) does not satisfy the condition $\phi = 0$ at $r \rightarrow \infty$. However, far away from the probe, the last term in the square bracket in (27) is on the order of $(m/M)^{1/2}$ and can be neglected.

The method of (27) solution is essentially the same as (10). Starting far from the probe, the potential has been evaluated in small steps toward the probe until the condition sheath-plasma boundary is met. This determines the dimensionless probe radius R/l and the plasma potential V_{plasma} (see Fig. 1) at this boundary.

Let us write down the expressions for all three currents of the problem:

$$I_i = 4\pi e R^2 N_0 \sqrt{\frac{2T}{M}} A \left(\frac{R}{l} \right), \quad (28)$$

$$I_{emiss} = g_i 4\pi e R^2 N_0 \sqrt{\frac{2T}{M}} A\left(\frac{R}{l}\right), \quad (29)$$

$$I_{e,plasma} = 4\pi e R^2 N_0 \sqrt{\frac{T}{2\pi m}} \exp(\Phi_{probe}). \quad (30)$$

Here, A is the previously defined function of the dimensionless probe radius. The condition of zero current to the probe yields the floating potential

$$\Phi_{float} = -\ln \left[A(1 + g_i) \sqrt{\frac{4\pi m}{M}} \right] = -\ln \frac{A}{1 - g_e} \sqrt{\frac{4\pi m}{M}}. \quad (31)$$

This potential includes the potential drops both in the sheath and inside the quasi-neutral plasma. Parameter A (dimensionless ion current) is, as before, a function of probe size (R/l) and also of the emission current fraction g_e .

IV. RESULTS OF CALCULATIONS FOR THE EMISSIVE PROBE

While solving Eq. (27), we noticed that at any given parameter B , the condition for the edge of the quasi-neutral plasma is satisfied at the same radius regardless of the emission level. This means that the relationship between the dimensionless ion current (parameter A) and the probe radius remains approximately the same as in the case of a non-emissive probe.

The following results were obtained using formula (15) for the approximate dependence of A on R/l (spherical probe) and formula (22) for a cylindrical probe.

Graphs for cylindrical and spherical probes coincide at the collisionless limit and then diverge as the ratio R/l increases. It can be seen in Fig. 8 that, as expected, the rise in emission current causes the floating potential to shrink. We didn't calculate for a very large g_e : when the sheath potential decreases below $(1-2)T_e$, the electron energy distribution declines from the Maxwellian one and the Boltzmann factor does not work at these low value barriers.

At the collisionless limit, the floating potential dependence on the emission current is the same for spherical and cylindrical probes. It is depicted in Fig. 9.

Increase in probe emission does not affect the quasi-neutral plasma, but shrinks the sheath, see Fig. 10

It was shown above that Bohm's criterion is satisfied at the sheath-plasma boundary of the non-emissive probes. Since the voltage drop in the quasi-neutral plasma practically does not depend on the emission level, one might expect that the same conclusion is correct for emissive probes also. Calculations proved that conclusion, Fig. 11.

Analyzing the above pictures, one can make the following conclusions:

1. As in the case of non-emissive probes, the float potential increases (becomes more negative) when the probe size changes from the collisionless regime toward the diffusion controlled one. The convenient approximate analytic formula for the floating potential is given.
2. Similar to the results of Hobbs and Wesson, as emission current ratio (g_e) approaches unity, the floating potential decreases down to about $-T/e$.
3. Calculations showed that the plasma potential at the plasma-sheath boundary is almost completely independent of the

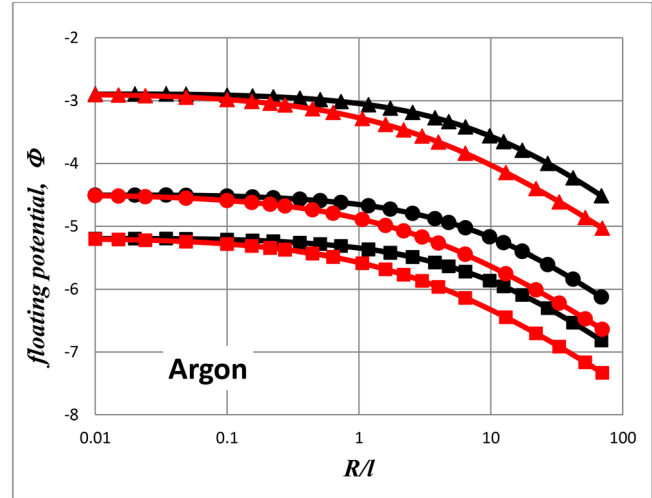


FIG. 8. Floating potential of emissive probe as function of probe size. Plasma gas: Argon. Different levels of emission current I_{emiss} to electron current from plasma $I_{e,plasma}$ ratio $g_e = I_{emiss}/I_{e,plasma}$ are shown. Triangles: $g_e = 0.9$; Circles: $g_e = 0.5$; Squares: $g_e = 0$. Black symbols: Cylindrical probe. Red symbols: Spherical probe.

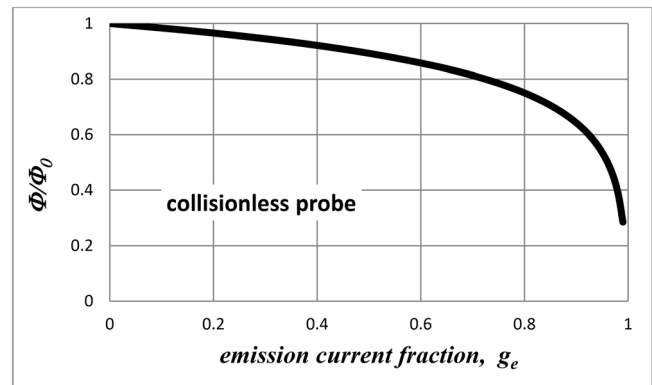


FIG. 9. Floating potential dependence on the emission current ratio. Collisionless limit. Φ_0 corresponds to non-emissive probe. Both cylindrical and spherical probes.

emission current. The conclusion that the ion velocity at that boundary corresponds to the Bohm criterion, obtained for non-emissive probes, also holds for emissive probes.

V. CONCLUSION

Ion currents to spherical and cylindrical probes immersed in plasma are calculated. The whole diapason of probe radius to ion free path ratio: from almost collisionless probes ($R/l \rightarrow 0$) to continuum probes ($R/l \rightarrow \infty$) is considered. Calculations were made for cold ions ($\theta = T_i/T_e = 0$) augmented by correction to include "warm" ions, assuming a small T_i/T_e ratio ($\theta = T_i/T_e < 0.2-0.3$), see Appendix B. In view of this assumption and because the Orbital Motion was neglected, we did not expand calculations to the higher θ ratio although at first glance the result looks reasonable even for $\theta = 1$. The results are

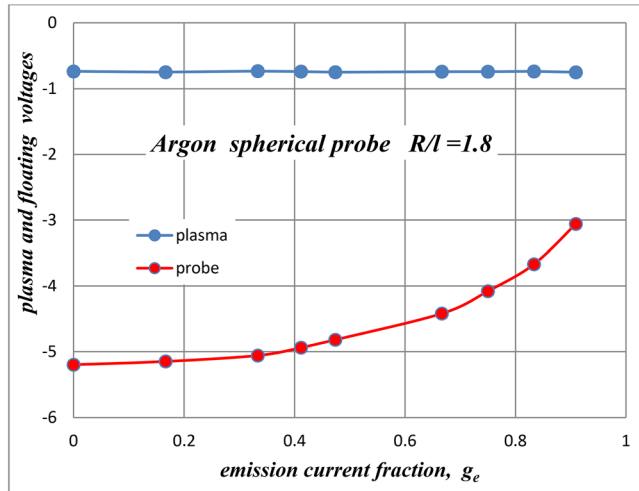


FIG. 10. Dimensionless voltage drop inside presheath plasma ϕ_{plasma} (blue line) and floating potential ϕ_{float} (red line).

presented in the form of simple formulas convenient for experimental data analysis.

For emissive probes, the floating potential under the assumption of cold ions was calculated for a wide range of probe radii: from collisionless up to diffusion controlled probes. As in the case of non-emissive probes, simple approximate formulas are presented to calculate the float potential. We do not solve the Poisson equation assuming monotonic potential distribution inside the sheath. Correspondingly, we did not consider transition to the space charge limited mode.

We would like to suggest experiments to check the results of the work. Perform measurements with two or three probes simultaneously: one, thin that satisfies condition $R/l < 1$ whereas another is a thick one that satisfies $R/l \sim 1$ or even $R/l > 1$. By changing the plasma density one can shuffle the whole range of R/l .

As for emissive probes, it is suggested to measure the float potential by two different methods: One by a non-emissive probe and another by an emissive probe and compare the results.

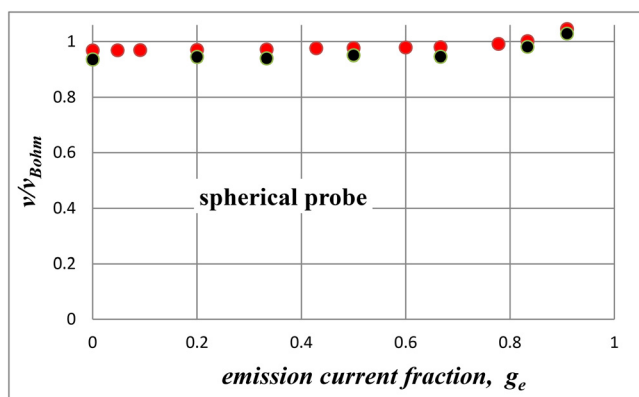


FIG. 11. Ion velocity at the sheath-presheath boundary. Red symbols: $R/l = 1.8$; black symbols: $R/l = 0.36$.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Valerian Nemchinsky: Investigation (equal); Methodology (equal); Supervision (equal); Validation (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

APPENDIX A

We replace the integral in (9) with Gauss–Laguerre integral sum³⁵

$$e^{\phi(\rho)} = \frac{B}{\rho^2} \sum_k C_k \sqrt{\frac{s_k}{\phi(\rho + s_k) - \phi(\rho)}}, \quad (\text{A1})$$

where C_k and s_k are weights and abscissas of the quadrature formula. Suppose, function ϕ is already known for radii larger than some radius ρ_0 . For a smaller radius, $\rho = \rho_0 - \Delta\rho$ such as $\Delta\rho < s_1$ (B1) is the equation for $\phi(\rho)$ at that smaller radius. Starting from a large radius, where the asymptote holds, it is possible to move step by step toward a smaller radius. This process repeats until the condition of the plasma-sheath boundary is met.

APPENDIX B

1. Warm ions correction for spherical non-emissive probes

One might see that the ion non-zero temperature correction is important for relatively large probes. In the case of a probe with a radius of one or two ion paths, the condition of a strong electric field exists inside the entire disturbed region. In this case, when the potential drop on a free path is on the order of T_e , ion velocity just after collision is not important. Another situation is for probes with $R \gg l$ when in the major part of the disturbed plasma, the potential drop on an ion free path is much less than T_e and comparable to T_i . The condition of the strong electric field still holds only at a few ion free paths close to the probe.

It is worthwhile to separate the probe proximity region into two: The region of a strong electric field $E l \gg T_i$ (E is the electric field) where the proposed above model is applicable and the region of a weak electric field. In the latter region, ion motion is described by ambipolar diffusion taking into account the finite ion temperature. The boundary between these two regions takes place where $El = \gamma T_i$, where γ is a number on order of unity.

In region II, one has

$$\frac{I_i}{4\pi r^2} = -D_a \frac{dN}{dr}, \quad (\text{B1})$$

where D_a is the coefficient of ambipolar diffusion. Since $N = N_0 \exp(\phi)$, keeping the linear terms, we obtain

$$\frac{I_i}{4\pi r^2} = -D_a N_0 e^{\phi} \frac{d\phi}{dr} \text{ i.e. } \approx \frac{D_a N_0}{T_e} E. \quad (\text{B2})$$

Thus, the condition at the boundary yields

$$\frac{I_i l}{4\pi r_c^2 D_a N_0} = \gamma \frac{T_i}{T_e}, \quad (\text{B3})$$

where r_c is the radius of the weak-strong fields boundary. The potential drop inside the region of a weak field is

$$\phi_c = - \left[\gamma \frac{T_i}{T_e} \frac{I_i}{4\pi l D_a N_0} \right]^{1/2}. \quad (\text{B4})$$

Let us integrate Eq. (12) (in the main text), describing potential distribution in the region of a strong field, from the critical radius r_c to some (yet unknown) the probe radius

$$e^{2\phi_c} - e^{2\phi} = \frac{(B\pi)^2}{6} [\rho^{-3} - \rho_c^{-3}]. \quad (\text{B5})$$

The terms $\exp(2\phi_c)$ and ρ_c^{-3} have different orders of magnitude. The first is proportional to small parameter $(T_i/T_e)^{1/2}$ whereas the second is proportional to $(T_i/T_e)^{3/2}$ and, therefore, can be dropped. We have

$$\phi = \frac{1}{2} \ln \left[\exp(2\phi_c) - \frac{(B\pi)^2}{6\rho^3} \right] \approx \frac{1}{2} \ln \left[1 + 2\phi_c - \frac{(B\pi)^2}{6\rho^3} \right]. \quad (\text{B6})$$

This potential trends to infinity (probe radius) when

$$B = \frac{1}{\pi} [6\rho^3 (1 + 2\phi_c)]^{1/2} \approx \frac{1}{\pi} \sqrt{6\rho^3 (1 + \phi_c)}. \quad (\text{B7})$$

We are looking for the parameter $\frac{l}{N_0}$ that is used to measure plasma density. Parameter B is proportional to it. Keeping in mind that ϕ_c also depends on B, Eq. (B7) is, therefore, the equation for B.

Let us put in (B4), the B value obtained in the cold ions approximation. We have $\phi_c = - \left[\gamma \frac{3\sqrt{6}}{\pi} \left(\frac{T_i}{T_e} \right)^{1/2} \left(\frac{R}{l} \right)^{3/2} \frac{1}{1 + T_e/T_i} \right]^{1/2}$ or, after redefining the numerical coefficient γ

$$\phi_c = - \left[\gamma \left(\frac{T_i}{T_e} \right)^{1/2} \left(\frac{R}{l} \right)^{3/2} \frac{1}{1 + T_e/T_i} \right]^{1/2}. \quad (\text{B8})$$

After putting this into (B7) we have expression for B and finally for $A = B(l/R)^2$

$$A = \frac{\sqrt{6}}{\pi} \left(\frac{R}{l} \right)^{-1/2} \left\{ 1 - \left[\gamma \left(\frac{T_i}{T_e} \right)^{1/2} \left(\frac{R}{l} \right)^{3/2} \frac{1}{1 + T_e/T_i} \right]^{1/2} \right\}. \quad (\text{B9})$$

Note that the correction factor is close to unity for probes with $R/l \leq 1$. Therefore, one can write

$$A = A_0 \left(\frac{R}{l} \right) \left\{ 1 - \left[\gamma \left(\frac{T_i}{T_e} \right)^{1/2} \left(\frac{R}{l} \right)^{3/2} \frac{1}{1 + T_e/T_i} \right]^{1/2} \right\}, \quad (\text{B10})$$

where $A_0(R/l)$ is the corresponding function describing the $T_i = 0$ case. For example, one can use (15), main text approximation.

It is worthwhile to find the coefficient γ based on the condition of a smooth match of the above correction and the known ion current dependence on probe radii at large R/l .

The dependence of dimensionless ion current A on the probe radius for large radii can be easily obtained. The ions move to the probe by means of ambipolar diffusion.

Integration (B1) from some radius to infinity gives

$$\phi(r) = \ln \left(1 - \frac{I_i}{4\pi r N_0 D_a} \right). \quad (\text{B11})$$

From this equation, one obtains the relationship between ion current and density of the undisturbed plasma

$$I_i = 4\pi R D_a N_0. \quad (\text{B12})$$

Formula (B12) coincides with formula (1.23) of the work by Su and Lam.⁵ For the ambipolar diffusion coefficient, one has $D_a = \frac{1}{3} (1 + T_e/T_i) l \sqrt{2T_i/M}$. Thus, for large probe radii the parameter A takes the form

$$\begin{aligned} A &= \frac{D_a}{R \sqrt{2T_e/M}} = \frac{1}{3} \frac{(1 + T_e/T_i) l \sqrt{2T_i/M}}{R \sqrt{2T_e/M}} \\ &= \frac{1}{3} \left(\frac{R}{l} \right)^{-1} \left[\sqrt{\frac{T_e}{T_i}} + \sqrt{\frac{T_i}{T_e}} \right]. \end{aligned} \quad (\text{B13})$$

The procedure for finding the value of γ is demonstrated in Fig. 12.

This procedure was applied to find γ values for $\theta = T_e/T_i$ inside the interval from 3 to 30. The results are shown in Fig. 13. The transition from the strong field (although with correction) mode to the fully weak field (diffusion) mode can be seen. The larger T_e/T_i ratio, the larger range of strong field regimes.

Figure 14 shows γ values that provide the best strong field-weak field match

APPENDIX C

Here, we compile the suggested approximate formulas scattered around the paper:

Cylindrical probe, cold ions:

$$I_i = A_c(R/l) \times 2\pi e N_0 R L \sqrt{\frac{2T}{M}},$$

$$A_c \left(\frac{R}{l} \right) = \frac{0.43}{1 + 0.48 (R/l)^{0.65}}.$$

Spherical probe, cold ions:

$$I_i = A_s(R/l) \times 4\pi e N_0 R^2 \sqrt{\frac{2T}{M}},$$

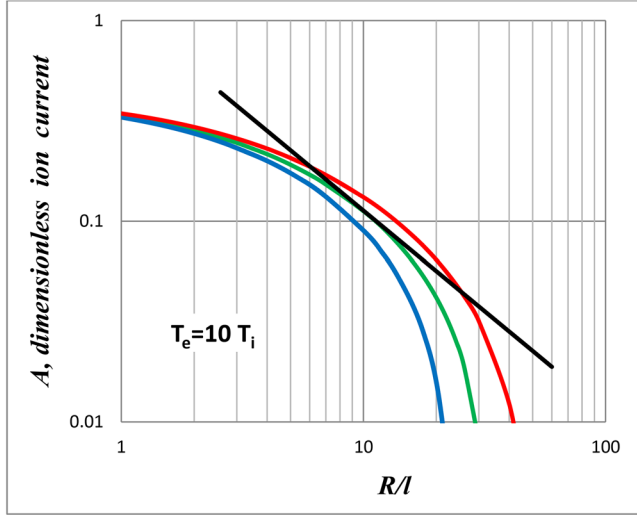


FIG. 12. Choosing the proper value of the number γ for $T_e/T_i = 10$. The colored lines obtained with (B10). The red line: $\gamma = 0.1$; the blue one – with $\gamma = 0.3$. The black line represents $A(R/l)$ dependence (B13). The smooth transition takes place at $R/l = 9$ when $\gamma = 0.18$ (the green line). At $R/l < 9$, A is represented by the green line, at $R/l > 9$, by the black one.

$$A_s\left(\frac{R}{l}\right) = \frac{0.43}{1 + 0.18(R/l)^{0.85}}.$$

Voltage drop at the sheath-presheath boundary for both spherical and cylindrical probes

$$\frac{eV_{\text{plasma}}}{T} = -\ln(\sqrt{2}A).$$

Spherical probe, warm ions:

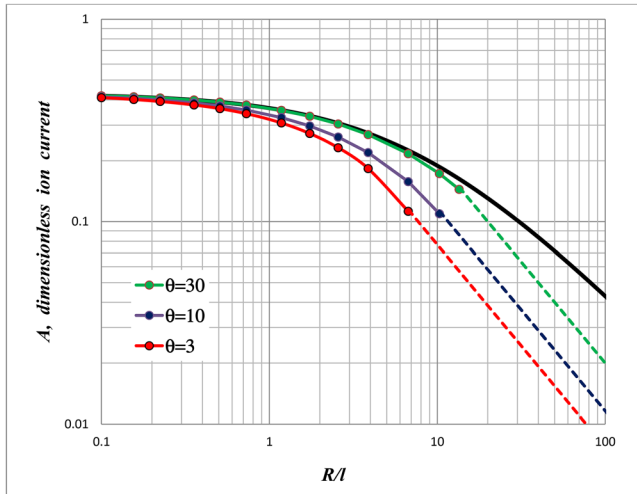


FIG. 13. $A = \frac{l}{4\pi n_0 R^2} \sqrt{\frac{M}{2T_e}}$ as function of probe R/l at different $\theta = T_e/T_i$. Solid colored line sections are calculated according to (B10), dashed colored line sections to (B13). Black line represents zero ion temperature approximation.

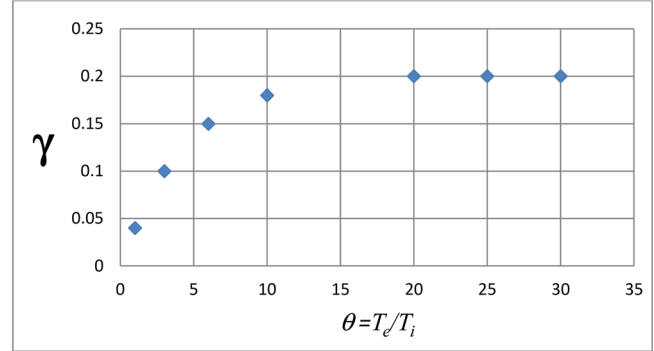


FIG. 14. Matching parameter providing smooth transition at the strong field-weak field boundary.

Parameter γ should be chosen for the best matching with the strong field dependence function $A(R/l)$

$$A_{\text{strong}}\left(\frac{R}{l}\right) = A_s\left(\frac{R}{l}\right) \left\{ 1 - \left[\gamma \left(\frac{T_i}{T_e} \right)^{1/2} \left(\frac{R}{l} \right)^{3/2} \frac{1}{1 + T_e/T_i} \right]^{1/2} \right\}$$

and weak field dependence $A_{\text{weak}}\left(\frac{R}{l}\right) = \frac{1}{3} \left(\frac{R}{l} \right)^{-1} \left[\sqrt{\frac{T_e}{T_i}} + \sqrt{\frac{T_i}{T_e}} \right]$.
Emissive probe, cold ions. Floating potential:

$$\frac{eV_{\text{float}}}{T} = -\ln \frac{\sqrt{2}A\left(\frac{R}{l}\right)}{1 - g_e} \sqrt{\frac{2\pi m}{M}}$$

with functions either A_c or A_s depending on the probe geometry.

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